Meghana Thimmaiah

Chicago, IL | +1 (312) 731-8295 | meghana.thimmaiahn60@gmail.com | LinkedIn | GitHub

(Open to Relocate)

SUMMARY

Data Analyst with **2+** years of experience designing scalable data pipelines and delivering actionable insights across **Supply Chain**, **Logistics**, and **Automotive** domains. Proficient in **SQL**, **Python**, **Power BI**, **Databricks**, **PySpark**, and **AWS**, with strong expertise in **ETL** development, **data cleaning** and **validation**, data modeling, **KPI** development, dashboarding, automated reporting, and ad-hoc analysis. Proven ability to translate business requirements into governed, analytics- and **ML-ready datasets**, collaborate with stakeholders, support executive reporting, and enable reliable, data-driven decision-making at scale. **Actively seeking impactful opportunities** to contribute to data-driven initiatives at scale.

AWARDS AND CERTIFICATIONS

- Databricks Fundamentals Accreditation by Databricks
- Microsoft Certified: Data Analyst Associate with Power BI
- Digital Marketing Analytics Internship by Stukent

Associated with Continental

- Received Spot Award for building and optimizing **AWS**-based **ETL** pipelines and **Redshift** analytics models, enabling KPI-driven operational and delivery risk reporting across multiple automotive programs.

Associated with GAINSystems

- Bravo Award - Recognized for consolidating multiple project workspaces into a single Databricks workspace, improving governance, collaboration, and operational efficiency.

PROFESSIONAL EXPERIENCE

GAINSYSTEMS — **Chicago, IL**

Data Analyst — **Jan 2025 - Present**

Customers: Rexel Group, Border States Electric, Amercare Royal, TW Metals, Colony Hardware, Reece Group

Delivered scalable, data-driven supply-chain forecasting and analytics by building production-grade ML pipelines, governed data models, and cloud-native ETL solutions on Databricks.

Project 1: Demand Data Engineering & ETL Platform

Enterprise customers provided raw demand transaction data from multiple ERP systems in file-based formats, but the data was inconsistent, unclean, and not directly usable for forecasting. I built a scalable Databricks-based ETL framework to transform this raw demand data into clean, standardized, ML-ready datasets used for demand forecasting.

- Designed and implemented Databricks **ETL pipelines** using **PySpark** and **SQL** to ingest customer demand data from CSV/Parquet files into Bronze, Silver, and Gold Delta tables, following the medallion architecture
- Built data cleansing, standardization, and aggregation logic to transform raw demand transactions into monthly, item-location–level, **ML**-ready datasets, enforcing business rules and data quality checks
- Experienced in the **R** programming language, proficient in statistical analysis, data visualization, and machine learning. Skilled in manipulating data frames, implementing statistical models, and creating visualizations using packages like **ggplot2**
- Built interactive **Tableau** dashboards to visualize KPIs, trends, and operational metrics for business and technical stakeholders
- Parameterized and scheduled recurring **Databricks** Jobs to support multiple enterprise customers using a single reusable **ETL framework** with customer-specific configurations, enabling reliable and repeatable demand data processing

Project 2: Explainable Demand Forecasting (SHAP & ML Pipelines)

Forecasting models worked well, but their predictions were not easy for planners to interpret. I added an explainability layer that analyzed model outputs and showed which features drove demand predictions, making the results transparent and usable for business teams.

- Implemented a Databricks-based **SHAP** explainability pipeline to compute feature-level contributions for monthly SKUL demand predictions, **improving transparency and business trust in ML forecasts**

- Integrated **MLflow** model artifacts, Gold Delta prediction tables, and feature datasets to generate SHAP values, visualizations (feature importance, contribution metrics), and explainability outputs for planners and analysts
- Used SQLAlchemy and **SQL** to query prediction data, feature metadata, and model outputs, and to persist SHAP results and summaries for downstream reporting, validation, and customer-facing analysis

Additional Responsibilities

- Managed **ML** demand forecasting pipelines to predict monthly SKUL demand, including dataset creation, model training, evaluation, and batch predictions
- Worked on **Gaussian Process Regression (GPR)** and probabilistic modeling to estimate prediction uncertainty and confidence intervals for lead times
- Performed **EDA** and data quality analysis using Pandas, Sweetviz, and YData Profiling to detect duplicates, missing values, and data issues before training
- Suggested improvements to data workflows and reporting processes based on data analysis, **reducing manual effort and recurring data issues**
- Managed secure access to data and services using Databricks Secret Scopes and **Azure** Key Vault, ensuring no credentials were hard-coded

CONTINENTAL ENGINEERING SERVICES — **Bangalore, India**
Data Analyst/ Data Engineer — **Feb 2023 - Dec 2023**

Delivered enterprise-scale operational and delivery analytics by centralizing project data through AWS-based ETL pipelines and performance-optimized Redshift reporting models.

Project 1: Enterprise Operational & Delivery Risk Analytics Platform

Continental managed hundreds of parallel automotive engineering programs, each maintaining separate risk registers with inconsistent tracking and no enterprise-wide visibility. I centralized this fragmented data and built a KPI-driven analytics platform to standardize operational, delivery, and compliance risk monitoring across projects and leadership teams.

- Built a **KPI-driven** operational, delivery, and compliance risk analytics platform by transforming **94K+** risk records across **600+** automotive projects into analytics-ready datasets and executive **Power BI** dashboards
- Standardized severity–probability risk scoring, lifecycle tracking, and overdue mitigation detection to monitor project delivery delays, operational constraints, and compliance gaps across the enterprise
- Enabled leadership and program managers to proactively identify high-risk projects, recurring delivery issues, and mitigation delays through portfolio-level views, historical trend analysis, and project-level drill-downs

Project 2: Enterprise Operational & Delivery Risk Analytics Platform

Automotive project, milestone, resource, and effort data were distributed across multiple internal systems and maintained through manual Excel-based reporting, leading to inconsistent metrics and delays in leadership reviews. I built and maintained an automated ETL pipeline to centralize this operational data, ensure data quality, and enable reliable reporting across automotive OEM programs.

- Designed and maintained end-to-end **ETL pipelines** using **Python** and **SQL** to ingest structured project, milestone, and resource data from multiple internal sources into **AWS S3** for centralized storage and analytics
- Implemented data transformation, standardization, and validation logic to clean duplicates, enforce business rules, and enrich datasets before loading them into Amazon Redshift, ensuring analytics-ready and trusted data
- Optimized Amazon **Redshift** reporting tables and **SQL** views using project-centric distribution keys, date-based sort keys, and pre-aggregated views to enable fast, reliable Power BI dashboards across multiple automotive OEM programs

Firm / Team Development Activities

- Built **KNIME** workflows to automate delivery KPI preparation (on-time delivery, utilization, effort variance) for an automotive OEM program, replacing manual Excel-based reporting and improving repeatability and accuracy
- Managed **Jira** workflows to track ETL tasks, data quality issues, and delivery milestones across automotive programs, improving cross-team visibility and coordination between engineering and analytics teams
- Supported operational decision-making by enabling consistent KPI definitions and reporting processes across programs, reducing ad-hoc reporting effort, and improving readiness for leadership reviews

BOSCH GLOBAL SOFTWARE TECHNOLOGIES **Bangalore, India**
Data Engineer/ Data Analyst **May 2020 - Jan 2023**
Customers: Hyundai Motor Group (HMG), Volkswagen, United Automotive Electronic Systems (UAES)
Built and maintained data pipelines, analytics workflows, and compliance reporting solutions using Python, SQL, and BI tools to support engineering, quality, and process teams across multiple automotive programs.

Project 1: ECU Test Data Consolidation & Analysis
ECU testing produced large volumes of diagnostic logs, CAN data, and test execution results that were spread across multiple tools, with each team cleaning and analyzing the data differently. I centralized and standardized this data to make the analysis consistent and easier to use.

- Parsed, cleaned, and normalized ECU diagnostic logs, CAN communication data, and test execution outputs using **Python, SQL,** and **Excel,** converting raw test data into analytics-ready datasets
- Aggregated and analyzed test results across multiple builds and ECU variants to identify recurring failures, regression issues, and high-risk components
- Integrated data from test execution tools, defect tracking systems, and requirement traceability platforms to correlate failures with software changes and coverage gaps
- Built interactive **Power BI** dashboards to visualize defect density, failure rates, test coverage, and stability trends for test leads and program managers

Project 2: ASPICE Compliance Data Integration & ETL Pipeline Maintenance
ASPICE process data was spread across multiple tools and spreadsheets, making compliance tracking inconsistent and manual, especially when onboarding new projects. I maintained a shared ETL pipeline to centralize, clean, and validate ASPICE data, and extended it to onboard a new project while keeping metrics consistent and audit-ready across existing projects.

- Maintained a shared **ETL pipeline** using Python and SQL to ingest ASPICE data (requirements, test cases, traceability, reviews) into an on-prem SQL Server data warehouse
- Extended the pipeline to onboard a new project by adding configuration-driven ASPICE artifacts and mappings, without impacting four existing projects
- Standardized and validated ASPICE compliance data (traceability, coverage, review completeness), flagging non-compliant records before consumption by **Power BI** dashboards

Project 3: ETL Job Monitoring & Failure Handling Framework
ETL jobs across multiple Bosch projects were running on schedules, but failures were detected late and often required manual investigation. I helped add monitoring and error-handling so issues could be identified and fixed faster.

- Added logging and status tracking to scheduled **Python-** and **SQL**-based batch ETL jobs to capture execution start, completion, and failure details. Implemented basic retry logic and error summaries to speed up issue diagnosis, created job-status tables and simple SQL Server reports to monitor ETL health, and improved operational visibility across projects

Other Duties

- Created and maintained project-level ETL documentation (data flow diagrams, schemas, runbooks), reducing onboarding time for new team members by ~30% and improving consistency across shared pipelines
- Supported onboarding and mentoring of 2 junior engineers, assisting with pipeline walkthroughs, script reviews, and issue resolution, which reduced recurring support questions and rework
- Maintained and supported an internal office meeting room booking tool built using R Shiny, handling minor enhancements, bug fixes, and data updates to ensure smooth day-to-day usage by teams

Software Engineer **Aug 2018 - April 2020**
Customers: Fiat, Ford, Tata Motors
Performed Python-driven test automation, functional testing, and HIL validation to improve test coverage, repeatability, and system reliability across ECU programs.

- Executed **BFT, FT,** and **HIL** testing on automotive **ECUs** using test automation frameworks to validate end-to-end system behavior under real-vehicle scenarios
- Automated regression and system-level test cases using Python scripts, and performed debugging and fault analysis with **UDE** and **TRACE32** to improve test reliability and reduce manual effort

PROJECTS

Healthcare Data Engineering & Fraud Detection

- Developed an AWS Glue -> Snowflake ETL pipeline processing 2M+ healthcare claim records, enabling fraud detection and cost savings of 15%.
- Created Python and SQL anomaly-detection models to uncover fraudulent billing patterns and reduce false claims.
- Enforced **HIPAA** compliance through RBAC, encryption, and detailed audit logging.

Cloud-Based Data Warehouse For Sales Analytics

- Transitioned an on-prem **SQL** system to **AWS Redshift,** improving storage efficiency by 30% via schema refactoring and compression.
- Scheduled and maintained daily data flows with Airflow and Python, supporting dynamic BI dashboards.
- Designed star schemas and utilized Redshift Spectrum to query external S3 datasets.

Distributed Data Pipeline for Big Data Processing

- Devised a fault-tolerant ETL pipeline with Hadoop, PySpark, and Dataproc to process multi-terabyte datasets 35% faster.
- Orchestrated ingestion and transformation with Airflow and loaded outputs to BigQuery for near real-time analytics.
- Applied schema validation and data profiling to ensure reliability and traceability across data flows.

SKILLS

- **Data Engineering & ETL:** ETL Pipelines (Apache Airflow, AWS Glue, Talend), Data Warehousing & Modeling (Redshift, Snowflake, BigQuery), Data Lakes & Processing (AWS S3, Databricks, Spark, Hadoop).
- **Big Data & Cloud Technologies:** AWS (S3, Redshift, Lambda, EMR, Kinesis, Firehose, IAM), Google Cloud (BigQuery, Dataproc), Databricks, Hadoop, Spark (PySpark, Spark SQL), Kafka, Hive.
- **Programming & Databases:** SQL (PostgreSQL, MySQL, Microsoft SQL Server), Python (Pandas, NumPy, PySpark, SQLAlchemy, Scikit-learn), R programming, C, Java, MongoDB, Snowflake, Cassandra, DynamoDB.
- **Business Intelligence & Data Governance:** Power BI, Tableau, Looker, Data Governance & Performance Optimization, Data Security & Compliance: Role-based access control, IAM.

EDUCATION

ILLINOIS INSTITUTE OF TECHNOLOGY — **Chicago, IL**
Master of Science in Computer Science (Concentration: Data Science) | GPA: 3.2/4 — **Jan 2024 - Dec 2025**

BMS INSTITUTE OF TECHNOLOGY & MANAGEMENT — **Bangalore, India**
B.E Electronics & Communication Eng — **July 2014 - May 2018**